

Does Idiomatic Phrasing Degrade LLM Task Performance?

A Controlled Study Across Instruction-Tuned Decoder Models

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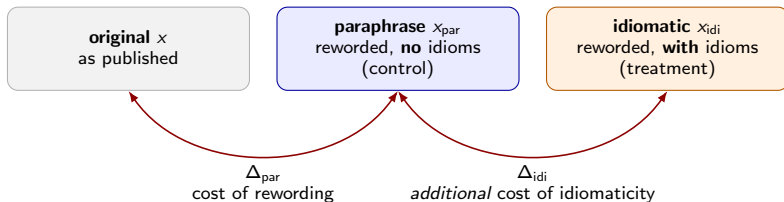
The Problem

- Idioms are non-compositional: a model must have *learned* an expression, not derived it.
- But any two rewordings already differ lexically and syntactically.
- So: does an accuracy drop after “idiomatizing” a sentence reflect **idiomaticity**, or just generic **rewording noise**?

Research question: does idiomatic phrasing cost accuracy *over and above* what any comparable non-idiomatic paraphrase already costs?

Design: A Paraphrase-Controlled Rewrite

Every original example x becomes two matched rewrites sharing a rewriting budget and differing in one property. The label y is given to the rewriter and re-checked afterwards.



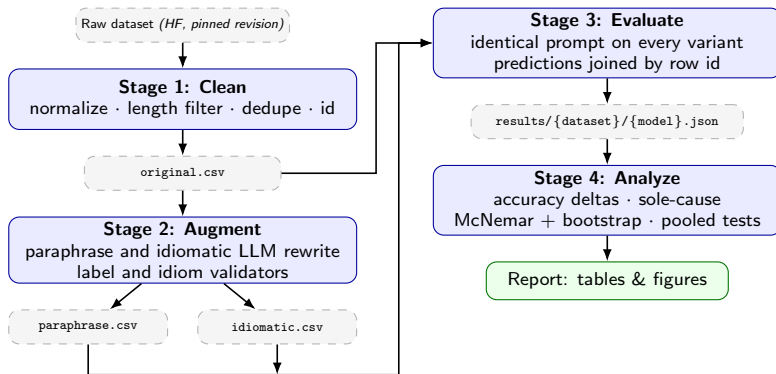
$$\Delta_{\text{par}} = \text{acc}_{\text{par}} - \text{acc}_{\text{orig}} \quad \Delta_{\text{idi}} = \text{acc}_{\text{idi}} - \text{acc}_{\text{par}}$$

Δ_{idi} is measured against **paraphrase**, not original, so the rewording cost is subtracted out.

Why a Control Arm Is Necessary

- Models are known to be sensitive to any semantics-preserving rewording (SEARs; Llama-2-70B-chat varying by 45.5 points between the best and worst of a set of semantically equivalent queries).
- A bare original-vs-idiomatic contrast confounds idiomaticity with generic paraphrase sensitivity.
- Prior idiom work measures *comprehension* (detection tasks), not the downstream cost on a task not about idioms.
- Our design is a matched, semantics-preserving control arm over the *same* items, not previously done for idiom-inserted NLU rewrites.

Pipeline



Four independently re-runnable stages, communicating only through versioned files on disk. `original.csv` feeds both Stage 2, as the rewriting source, and Stage 3, as the control variant.

Datasets & Models

Three datasets, one rewritten field each (in **bold**):

- **SST-2** (binary classification): label one **sentence** *positive* or *negative*.
- **MMLU** (4-way multiple choice): answer a knowledge **question stem** with A–D; the four choices stay as published.
- **MNLI** (3-way inference): given a **premise** and a hypothesis, answer *entailment*, *neutral* or *contradiction*; the hypothesis stays as published.
- Aligned triples: 1,548, 1,085 and 4,709 rows

14 instruction-tuned decoder LLMs, 0.5B–7B, 10 families (Qwen2.5, Falcon3, OLMo-2, Gemma-2, Mistral, Phi-4-mini, Granite-3.1, H2O-Danube3, Yi-1.5, SmolLM2), zero-shot under an identical prompt per dataset.

42 (model, dataset) runs, so 126 variant-level inference passes.

Statistical Framework

Sole-cause test: restrict to rows the model got right on *original*, so any later error is caused by the rewrite, then count b = idiomatic-only-wrong, c = paraphrase-only-wrong. Rows where both rewrites agree carry no information and drop out.

Tested two ways, required to *agree*:

- **McNemar** χ^2 (*parametric*) with the Edwards continuity correction:

$$\chi^2 = \frac{(|b - c| - 1)^2}{b + c}, \quad \text{df} = 1$$

null: among discordant rows it is a coin flip which rewrite broke it.

- **Paired bootstrap** (*non-parametric*), 10,000 resamples of $b - c$, assuming no distribution, since the χ^2 approximation is unreliable when $b + c$ is small.

The parametric and non-parametric verdicts **agree in all 42 per-model runs**, so no reported result depends on the χ^2 approximation holding.

Results: One Manipulation, Three Directions

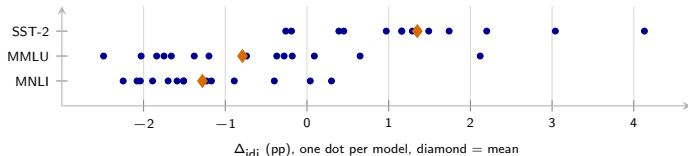
Dataset	n	Δ_{par}	Δ_{idi}	sig. (-/+)
SST-2	1,548	-1.28	+ 1.35	0/5
MMLU	1,085	+0.13	-0.79	1/0
MNLI	4,709	-0.68	- 1.28	10/0

- **MNLI**: significant penalty for 10 of 14 models, and the sole-cause test agrees ($b > c$ for 12 of 14).
- **SST-2**: idioms *help*, significant for 5 models and negative for none. One reading is that idioms here carry sentiment rather than obscure it, but nothing we ran isolates that.
- **MMLU**: marginal, surviving only in aggregate (1 of 14 individually, $p = 0.0011$ pooled).

Not a general accuracy tax: the effect flips *sign* across tasks.

Every Model, Every Dataset

Model	SST-2	MMLU	MNLI	Model	SST-2	MMLU	MNLI
Qwen2.5-0.5B	+0.45	−1.75	−0.40	Phi-4-mini-3.8B	−0.19	−1.84	− 1.51 [†]
OLMo-2-1B	+1.29	−0.28	− 1.70 [†]	H2O-Danube3-4B	+ 1.74	− 2.49	+0.04
Qwen2.5-1.5B	+1.16	−2.03 [†]	− 1.59 [†]	Yi-1.5-6B	+0.97	−0.18	− 1.89 [†]
SmolLM2-1.7B	+ 2.20	−0.74	+0.30	Qwen2.5-7B	+ 3.04 [†]	+0.09	− 1.17 [†]
Gemma-2-2B	+1.36 [†]	−1.38	− 1.51 [†]	Falcon3-7B	−0.26	+0.65	− 1.23 [†]
Granite-3.1-2B	+ 1.49 [†]	+2.12	−0.89	Mistral-7B-v0.3	+ 4.13 [†]	−1.20	− 2.08
Falcon3-3B	+0.39	−0.37	− 2.25 [†]	OLMo-2-7B	+1.16	−1.66 [†]	− 2.04 [†]



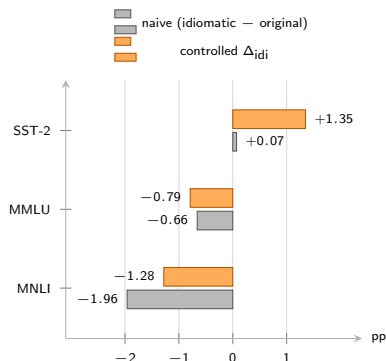
Ordered by parameter count. **Bold:** Δ_{idi} itself significant ($p < 0.05$, McNemar exact). [†]: the separate sole-cause b vs. c gap is significant on *both* tests. The two marks are independent, so a cell may carry either, both or neither.

Worked Disagreement Cases

Variant	User prompt	Prediction
MNLI , Falcon3-3B ($\Delta_{\text{idi}} = -2.25$ pp). Fixed hypothesis: She now knows what was going on in her mind. Gold: <i>entailment</i> .		
original	Premise: It took me really a long time to sort what was going on in my mind out , she said.	entailment ✓
paraphrase	Premise: She stated that it required a significant amount of time for her to organize her thoughts .	entailment ✓
idiomatic	Premise: It took me quite a while to wrap my head around what was going on in my mind, she said.	contradiction ×
SST-2 , Mistral-7B-v0.3 ($\Delta_{\text{idi}} = +4.13$ pp). Gold: <i>positive</i> .		
original	Sentence: is n't embarrassed to make you reach for the tissues	positive ✓
paraphrase	Sentence: is not ashamed to make you cry	negative ×
idiomatic	Sentence: isn't shy about pulling at your heartstrings and making you reach for the tissues	positive ✓
MMLU , H2O-Danube3-4B ($\Delta_{\text{idi}} = -2.49$ pp). The four choices are untouched: A. a logical mistake. B. an ethical mistake. C. a factual mistake. D. not a mistake. Gold: <i>D</i> .		
original	Question: According to Nagel , the view that moral luck is paradoxical is:	D ✓
paraphrase	Question: Based on Nagel's analysis , the perspective that moral luck is a paradox is:	D ✓
idiomatic	Question: In Nagel's eyes , the view that moral luck is paradoxical is:	A ×

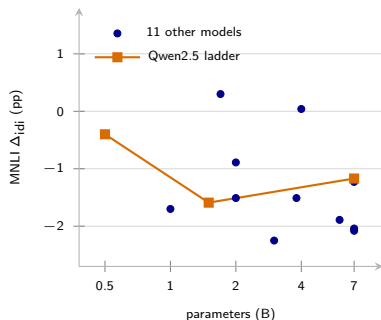
Each block is one row where a model with a *significant* Δ_{idi} split between the two rewrites. **Bold** marks the aligned span; only the rewritten field changes within a block.

The Control Arm Changes the Story



- Without a control arm, idiomatic can only be scored against *original*: SST-2 then reads as a null (+0.07 pp) and MNLI's cost is overstated by roughly 53%.
- The matched arm splits that false null into two real cancelling effects and cuts MNLI to its idiom-specific size.
- Modest magnitudes: MNLI's 1–2 pp is the same order as plain rewording's own cost, not an order above it.
- So idiomaticity is a **distinguishable contributor** to paraphrase sensitivity, not a separate failure mode.

Scale Does Not Explain It



- Across the 14 models, Δ_{idi} correlates with neither a model's parameter count nor its own accuracy on the original variant (Spearman per dataset, none significant): bigger and stronger models are not less affected.
- The Qwen2.5 ladder is non-monotone on MNLI (-0.40 , -1.59 , -1.17 pp at 0.5B/1.5B/7B); all four 7B models sit in the degraded group.
- That null is weak: with 14 points a Spearman test needs $|\rho| \geq 0.53$ to reach $p < 0.05$, and the largest of the six observed is 0.30. A moderate trend would go undetected.
- **Nothing supports waiting for a larger checkpoint**; settling it needs far more models.

Limitations

- Parse-failure filtering is blunt and non-random (up to 12.4% dropped for one model/dataset pair).
- Pooled tests treat repeated rows as independent, giving optimistic p-values. The fix is a mixed-effects model with a random effect per row, which we did not run: it would let a row that is idiom-hard for many models count once rather than 14 times, widening the pooled intervals. That falls hardest on MMLU, whose result rests on pooling alone.
- No multiple-comparisons correction: 14 tests at $\alpha = 0.05$ expect 0.7 hits under a true null, so MMLU's single significant model carries little weight.
- Zero-shot, greedy decoding, single prompt template, decoder-only panel.

Future Work

- Verify rewrites preserve meaning; run a mixed-effects re-analysis of the pooled claims.
- Correlate Δ_{id} with two per-idiom properties, rather than counting every idiom the same:
 - *opacity*: how far the idiom's meaning sits from its literal words (psycholinguistic norms). If the cost is non-compositionality, opaque idioms should cost more.
 - *frequency*: corpus counts, a proxy for pretraining exposure. If rarer idioms cost more, that is the familiarity account.

Both reuse the text already collected, but Stage 2 recorded no inventory of which idiom it inserted, so each row's idiom must be identified first.

- Rewrite MNLI's hypothesis or MMLU's answer choices too, making idiom position a manipulated variable *within* one dataset.
- Few-shot prompting: the cheap win is general accuracy and output format, cutting the parse failures that contaminate SST-2. The idiom penalty may not move at all.
- Brief fine-tuning on labelled idiomatic rewrites: the intervention aimed at the MNLI penalty itself, testing whether it is a fixable gap in exposure.

Summary

- 14 instruction-tuned decoder LLMs (0.5B–7B), 3 datasets, every idiomatic rewrite matched to a non-idiomatic paraphrase of the same row.
- The control arm is what makes the result readable: without it SST-2 reads as a null and MNLI is overstated by 53%.
- The penalty is **MNLI's**: -1.28 pp, 10 of 14 models. MMLU -0.79 (1 of 14); sentiment none. Means across models.
- Idiomaticity is a distinguishable contributor to paraphrase sensitivity, not a separate failure mode. No scale trend, but 14 models could not have found a moderate one.

Thank you

Questions?